Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | LinkedIn | GitHub | Portfolio

Summary

Full-stack software engineer with 2+ years building AI-enabled web applications, REST APIs, cloud services, and data workflows. Experienced in React, JavaScript, TypeScript, Python, Node.js, AWS, SQL/NoSQL, RAG, vector retrieval, CI/CD, testing, and production performance.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python APIs, Azure SQL services, and user-facing dashboards for 500+ users, tuning indexed queries and backend access paths to reduce production response latency from 1.2 seconds to 350 milliseconds.
- Implemented reliable workflows for 5,000+ monthly jobs using explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving system uptime, observability, maintainability, and production troubleshooting responsiveness.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React interfaces and Flask REST services handling 10,000+ daily requests, integrating secure document ingestion, authentication, status updates, and automated grading workflows while resolving production bugs for 150+ users.
- Created multithreaded processing for 1,000+ files with validation, AWS RDS persistence, progress reporting, and failure handling, reducing turnaround 60% while maintaining reliable application behavior under concurrent production load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built testable PyTorch pipelines processing 100,000+ trajectory logs, improving long-horizon planning accuracy 18% through reproducible experiments, structured evaluation, performance monitoring, and systematic debugging of intelligent model behavior.
- Developed a spatial-temporal encoder combining map geometry and agent context through multi-head attention, improving closed-loop stability 15% while documenting experiments and validating performance across diverse simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed Python services, SQL integrations, and ETL workflows using AWS Glue, S3, and REST APIs, improving response times 30% and data-processing efficiency 68% across production cloud applications.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating application audits, defect diagnosis, release validation, and reliable deployments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular React and Node.js features across eight microservices with authentication, role-based access, WebSocket messaging, and state synchronization, coordinating 2 million+ Azure SQL records through secure interfaces.
- Automated Docker builds, functional tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving software quality, deployment consistency, and operational visibility.

Skills

Frontend & Languages: JavaScript, TypeScript, Python, Java, SQL, HTML5, CSS3, React, Next.js, responsive interfaces, reusable components
Backend & AI: Node.js, Flask, FastAPI, REST APIs, microservices, WebSockets, LLaMA-2, RAG, FAISS, semantic search, vector databases, prompt engineering
Data & Cloud: PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, Firebase, AWS RDS, AWS, Azure, S3, Blob Storage, data modeling, query optimization
Quality & Delivery: Git, Docker, Kubernetes, Jenkins, CI/CD, PyTest, Mockito, automated testing, unit testing, integration testing, debugging, logging, monitoring, code review, documentation, production support, Agile

Projects

Big-Brain | Live | *LLaMA-2, RAG, FAISS, PostgreSQL, REST APIs, Clerk*
- Built an authenticated RAG application processing 50,000+ unstructured documents with LLaMA-2, FAISS, PostgreSQL, and REST APIs, combining semantic retrieval, conversational context, access controls, and real-time collaboration workflows.

HMIS | Live | *TypeScript, React, REST APIs, Appwrite, Sentry, Twilio*
- Built responsive healthcare workflows for appointments, records, and consent-based access using React, REST APIs, authentication, notifications, and Sentry monitoring, supporting 100+ daily interactions while reducing administrative workload 30%.

PiXMod | Live | *Next.js, TypeScript, MongoDB, Stripe, CLIP, SAM*
- Built a full-stack AI application using Next.js, TypeScript, CLIP, and SAM, implementing reusable interfaces, secure authentication, MongoDB persistence, Stripe integration, and responsive image workflows across multiple devices.